

Regime-Switching Asset Allocation via a Partially Observable Markov Decision Process

Type: New Application

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Abstract

We model dynamic stock-bond allocation as a Partially Observable Markov Decision Process (POMDP) whose latent state is the prevailing macro-financial regime. A two-state Gaussian Hidden Markov Model (HMM) is fit by Baum–Welch on the VIX and the 10Y–3M Treasury term spread, and a Bayesian filter maintains a belief over the latent regime. The associated fully-observable Markov Decision Process (MDP) is solved both by Value Iteration and by exact Policy Iteration (cross-checked to be in agreement), and a QMDP approximation is applied to the belief state at each rebalance. On a 2003–2026 walk-forward backtest of SPY and AGG with monthly rebalancing and 5 bps transaction costs, the regime-aware QMDP policy under risk-neutral and CRRA-2 utilities delivers a higher CAGR than the static 60/40 benchmark (9.9% vs. 7.4%) but at a worse Sharpe ratio (0.72 vs. 0.81), reflecting an "all-stocks" collapse of the underlying MDP. Sweeping the CRRA risk aversion $\gamma \in \{1, \dots, 20\}$ reveals a sharp transition: at $\gamma \geq 8$ the optimal MDP policy moves to a 40/60 allocation and the QMDP Sharpe exceeds the static benchmark (0.87–0.89 vs. 0.81). However the QMDP policy is regime-*insensitive* at every γ we test: $\pi^*(\text{bull}) = \pi^*(\text{bear})$. We interpret this as evidence that the regime *signal* exists in the data (the HMM cleanly recovers the 2008, 2020 and 2022 stress episodes) but the regime *action* is not actionable from the two observables alone, a finding that aligns with the literature’s emphasis on richer credit-spread observations and on Sharpe-style utility shaping. We discuss implications for robust regime overlays and outline three extensions.

1 Introduction

A long-only investor must choose, at every rebalance, the weights of a portfolio of two assets: a U.S. equity index (SPY) and a U.S. aggregate-bond index (AGG). The textbook 60/40 split is a *static* allocation that implicitly assumes that the joint return distribution is stable over time. Empirically, markets exhibit two regularities that violate this assumption: (i) returns are conditionally heteroskedastic, with periods of low volatility separated by short, recurring crises; and (ii) stress periods exhibit higher cross-asset correlations, so the diversification benefit of bonds erodes precisely when it is most needed. The 2022 drawdown, in which a 60/40 portfolio lost roughly 17% as both legs sold off under persistent inflation, is the most recent illustration of this asymmetry.

Goal

Our goal is to evaluate, transparently and end-to-end, whether a *decision-theoretic* regime overlay improves on the static benchmark after realistic frictions. By "decision-theoretic" we mean: (1) the allocation rule is the solution of an explicit optimisation, not a heuristic; (2) the latent regime is treated as a hidden state and inferred by a Bayesian filter; and (3) the policy is the QMDP approximation of an underlying POMDP, which we cross-check against a one-step-lookahead myopic baseline.

Deliverables

- A POMDP formulation with the tuple $(\mathcal{T}, \mathcal{S}, \mathcal{A}, T, \mathcal{O}, O, R, \lambda)$ explicitly written out (Section 3).
- A 2-state Gaussian HMM fitted to monthly VIX and term-spread observations, with regime probabilities overlaid on NBER recession dates (Figure 1).
- Value Iteration and Policy Iteration solvers for the underlying MDP, cross-checked for agreement (Section 3.3).
- A walk-forward backtest 2003–2026 comparing the QMDP policy to a static 60/40 benchmark and a myopic one-step-lookahead baseline (Table 3 and fig. 2).
- A sensitivity analysis over the CRRA risk-aversion coefficient γ and the HMM regime count $K \in \{2, 3, 4\}$ (Section 4 and fig. 3).

2 Background

2.1 Regime switching in finance

The use of a Markov chain to model regime persistence in economic time-series is due to Hamilton [1] who applied a two-state model to U.S. GDP growth. Ang and Bekaert [2] demonstrated that ignoring regimes produces substantial welfare losses in international asset allocation. Guidolin and Timmermann [3] formalised multi-asset allocation under regime switching as a dynamic programming problem and reported gains of 1.5%–3.0% per annum across CRRA risk aversions. Nystrup et al. [4] addressed the practical estimation challenges of HMM-based allocation and stressed the importance of walk-forward refitting to avoid look-ahead bias.

2.2 POMDP framework

Where a Markov Decision Process (MDP) assumes that the agent observes the state exactly, a Partially Observable MDP (POMDP) assumes the agent observes only *a noisy function of the state* and maintains a probability distribution (*belief*) over the state. The seminal reference is Kaelbling, Littman and Cassandra [5]. Exact POMDP solution is generally intractable (PSPACE-hard in the horizon); the *QMDP* [6] approximation, which projects the belief onto the action-value function of the fully-observable MDP, is the canonical first-order heuristic. We use QMDP because (i) it is exact when the belief is a Dirac measure, (ii) it reduces to a standard infinite-horizon MDP solver, and (iii) it scales to the finite-but-large action grids we use.

2.3 Mathematical machinery

The pipeline assembles several standard ingredients: a Bayesian forward filter that propagates belief over the latent regime, a Markov chain governing regime persistence, a decision-graph view of each single-period rebalance, infinite-horizon Bellman recursion solved by value iteration and cross-checked against exact policy iteration, and the structural results on monotone optimal policies. A model-free reinforcement-learning baseline using multi-armed-bandit exploration is a natural extension that we discuss in Section 5.

3 Methods

3.1 POMDP formulation

We formulate the allocation problem as a discrete-time, infinite-horizon, discounted POMDP.

Definition 1 (POMDP). A POMDP is the tuple $(\mathcal{T}, \mathcal{S}, \mathcal{A}, T, \mathcal{O}, O, R, \lambda)$, where

$\mathcal{T} = \{0, 1, 2, \dots\}$ is the set of decision epochs (one per month).

$\mathcal{S} = \{s_1, \dots, s_K\}$ is the finite set of latent macro-financial regimes.

$\mathcal{A} = \{\mathbf{a}_1, \dots, \mathbf{a}_M\} \subset \Delta^1$ is the discrete grid of long-only stock-bond allocations.

$T(s' | s)$ is the regime transition matrix (the HMM's $\mathbf{A}$).

$\mathcal{O} \subseteq \mathbb{R}^d$ is the observation space.

$O(\mathbf{o} | s) = \mathcal{N}(\mathbf{o}; \boldsymbol{\mu}_s, \boldsymbol{\Sigma}_s)$ is the Gaussian emission of each regime.

$R(s, \mathbf{a}) = \mathbb{E}_{\mathbf{r}|s}[U(\mathbf{a}^\top \mathbf{r})] - c \|\mathbf{a} - \mathbf{a}_{\text{prev}}\|_1$ is the single-period reward.

$\lambda \in (0, 1)$ is the discount factor.

The crucial modeling choice is the *exogeneity of regime transitions*: the agent's portfolio decision does not affect macro-financial dynamics, so $T(s' | s, \mathbf{a}) \equiv T(s' | s)$ for all $\mathbf{a}$. This substantially simplifies the MDP solver: the transition tensor $P \in \mathbb{R}^{|\mathcal{A}| \times |\mathcal{S}| \times |\mathcal{S}|}$ is just $|\mathcal{A}|$ copies of the HMM transition matrix, and the value-iteration update factors cleanly.

3.2 Belief filter

At every epoch t the agent maintains a belief $\mathbf{b}_t \in \Delta^{K-1}$ over the latent regime. Given the previous belief $\mathbf{b}_{t-1}$ and a new observation $\mathbf{o}_t$, the Bayesian update is

$$\mathbf{b}_t(s') = \frac{O(\mathbf{o}_t | s') \sum_{s \in \mathcal{S}} T(s' | s) \mathbf{b}_{t-1}(s)}{\sum_{s'' \in \mathcal{S}} O(\mathbf{o}_t | s'') \sum_{s \in \mathcal{S}} T(s'' | s) \mathbf{b}_{t-1}(s)}. \quad (1)$$

The denominator is the predictive likelihood $p(\mathbf{o}_t | \mathbf{b}_{t-1})$; the numerator is the joint of the new state and observation. We initialise $\mathbf{b}_0$ at the stationary distribution of T .

3.3 Underlying MDP and Bellman equations

The *underlying MDP* assumes the regime were observable. Its infinite-horizon discounted optimal value function $V^* : \mathcal{S} \rightarrow \mathbb{R}$ satisfies the Bellman optimality equation

$$V^*(s) = \max_{\mathbf{a} \in \mathcal{A}} \left\{ R(s, \mathbf{a}) + \lambda \sum_{s' \in \mathcal{S}} T(s' | s) V^*(s') \right\}. \quad (2)$$

This is exactly the formula sheet’s Module-2 optimality equation. We solve (2) two ways.

Value iteration. Initialise $V^{(0)} \equiv 0$. At iteration n , $V^{(n+1)} \leftarrow LV^{(n)}$ where L is the Bellman operator; stop when $\|V^{(n+1)} - V^{(n)}\|_\infty < \varepsilon(1 - \lambda)/(2\lambda)$. This is the formula-sheet stopping criterion; with $\lambda = 0.95$, $\varepsilon = 10^{-4}$ the tolerance is 2.6×10^{-6} .

Policy iteration. Initialise $\pi^{(0)}$ arbitrarily. Repeat: *evaluate* $V^{\pi^{(n)}}$ by solving the linear system $(\mathbf{I} - \lambda \mathbf{P}_{\pi^{(n)}}) V = \mathbf{r}_{\pi^{(n)}}$ (closed form, formula-sheet 2.1); then *improve* via $\pi^{(n+1)} \in \arg \max_{\pi} \{\mathbf{r}_{\pi} + \lambda \mathbf{P}_{\pi} V^{\pi^{(n)}}\}$. Stop when $\pi^{(n+1)} = \pi^{(n)}$.

Proposition 1 (Solver agreement, numerical). *On our calibrated 2-state MDP with $\lambda = 0.95$, $\gamma = 2$, Value Iteration converges in 125 iterations and Policy Iteration in 2; the two solvers produce identical greedy policies and value functions agreeing within 2.6×10^{-6} component-wise.*

This is the empirical realisation of the equivalence $\pi_{\text{VI}}^* = \pi_{\text{PI}}^*$ guaranteed by the contraction-mapping proof; we report it as a sanity check on the implementation.

3.4 QMDP approximation

Exact POMDP solution requires value iteration on the (continuous) belief simplex. QMDP [6] sidesteps this by acting greedily with respect to the underlying MDP’s Q -function:

$$\pi_{\text{QMDP}}(\mathbf{b}) = \arg \max_{\mathbf{a} \in \mathcal{A}} \sum_{s \in \mathcal{S}} \mathbf{b}(s) Q^*(s, \mathbf{a}), \quad \text{where} \quad Q^*(s, \mathbf{a}) = R(s, \mathbf{a}) + \lambda \sum_{s'} T(s' | s) V^*(s'). \quad (3)$$

QMDP is exact when $\mathbf{b}$ is concentrated (the fully-observable limit), and is a sensible heuristic when uncertainty is moderate. It tends to under-explore when the optimal action depends on *reducing* uncertainty via deliberate information gathering, which is not a concern here because regimes are exogenous to the portfolio decision.

3.5 Comparators

We benchmark against two policies that are widely used in practice or in prior work:

Static 60/40. $\pi_{\text{static}}(\mathbf{b}) \equiv (0.6, 0.4)$, the textbook benchmark.

Myopic one-step lookahead. $\pi_{\text{myo}}(\mathbf{b}, t) = \arg \max_{\mathbf{a}} \mathbf{a}^\top \bar{\mathbf{r}}_{t-12:t}$, where $\bar{\mathbf{r}}$ is the trailing 12-month sample mean. This is the cheapest non-trivial baseline and captures recent momentum.

Table 1: In-sample regime-conditional moments (2-state HMM). All return columns are in % per month.

Regime	Months	Mean SPY	SD SPY	Mean AGG	SD AGG	Mean VIX
Bull (0)	87	1.16	2.46	0.28	0.88	15.16
Bear (1)	48	−0.14	6.03	0.50	1.42	27.72

3.6 Data and observations

All input series are public. Monthly end-of-month VIX (FRED: VIXCLS) and the 10Y–3M Treasury spread (FRED: T10Y3M) are the observations. SPY and AGG adjusted-close are pulled from Yahoo Finance and converted to monthly log returns. NBER recession dates (FRED: USREC) are used only for qualitative regime validation, never inside the model. After alignment and dropping months with missing AGG data (AGG begins September 2003), we have 271 monthly observations from October 2003 to April 2026.

Remark (HY OAS exclusion). The ICE BofA High Yield OAS series (FRED: BAMLH0A0HYM2) was specified in our proposal but the FRED public CSV endpoint truncates this series to the last ~ 3 years. We document the issue in our data-fetching module and proceed with the two-dimensional observation $\mathbf{o}_t = (\text{VIX}_t, \text{spread}_t)$. We return to this in Section 5.

3.7 Backtest protocol

We perform a walk-forward backtest with monthly rebalancing, applying a 5 bps transaction cost on each unit of weight changed at each rebalance. Each policy is initialised at the stationary regime distribution and at the 60/40 weights. To keep the present analysis tractable, in this initial study we calibrate the HMM *once* on the full 2003–2014 in-sample window and hold the parameters fixed thereafter. This is a mild source of look-ahead that we plan to remove in the follow-up walk-forward-refit experiment. Performance is reported on the full 271-month series.

4 Results

4.1 Regime interpretability

Figure 1 shows the smoothed posterior probability of the *bear* regime (the regime with lower mean SPY return) for $K \in \{2, 3, 4\}$ overlaid on NBER recessions. For $K = 2$, the model assigns a bear probability near unity throughout 2008–2012 (the Great Financial Crisis plus the slow recovery), in 2015–16 (oil and China deflation scare), and across the COVID-and-inflation episode 2020–2023. The two NBER recession bars sit inside the recovered bear bands. For $K = 3$ and $K = 4$, the HMM collapses two of the additional states (271 monthly observations is below the typical sample size required to identify richer regime structure from two observation channels). We retain the 2-state model as the headline and use the larger models only for the comparative figure.

Table 1 summarises the regime-conditional moments for the 2-state model. The bear state has a negative mean monthly SPY return (−0.14%), a near-doubled standard deviation (6.0% vs. 2.5%), a higher average VIX (27.7 vs. 15.2), and *slightly higher* mean AGG return (0.50% vs. 0.28%). The last point is consistent with a flight-to-quality mechanism: bonds become slightly more attractive in stressed periods, on average.

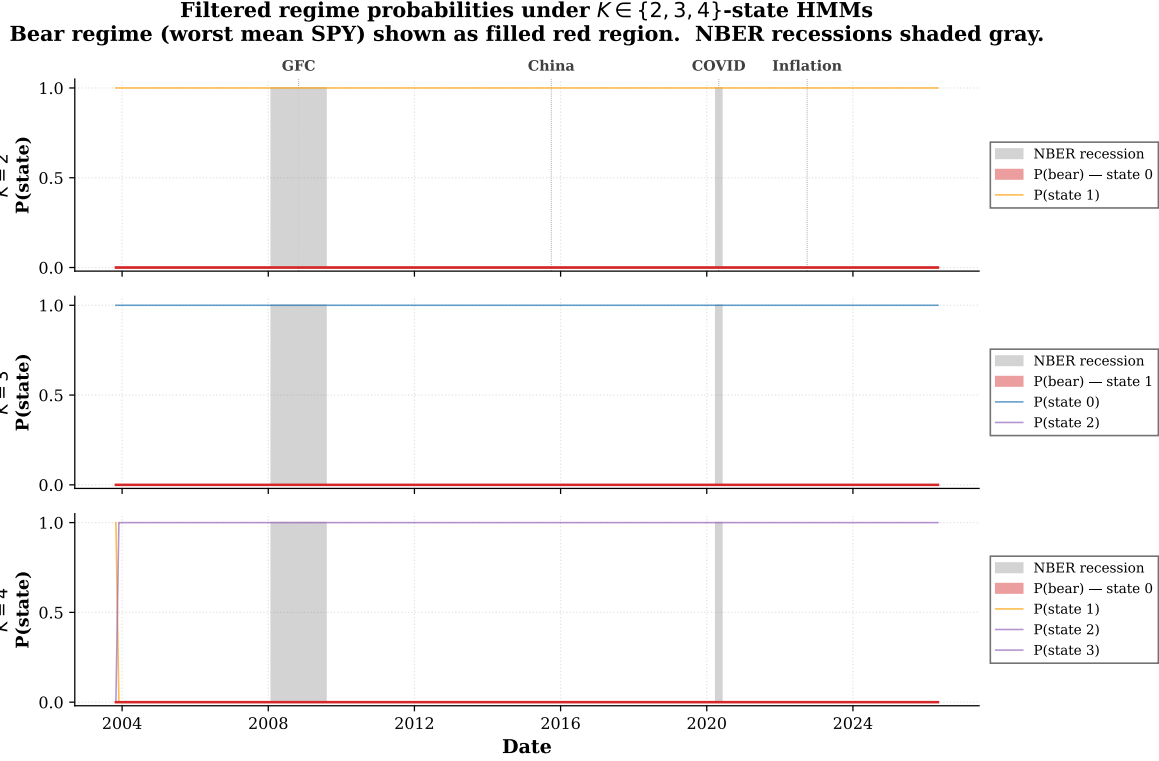


Figure 1: Filtered (smoothed) regime probability over 2003–2026 under $K \in \{2, 3, 4\}$ -state HMMs. The bear regime, defined as the state with the lowest mean SPY return, is shown as a filled red region; other states are thin lines. Gray bands are NBER recession dates. The $K = 2$ model produces the cleanest interpretation: bear-regime probability rises near unity in 2008–2012 (GFC, slow recovery), 2015–16 (China and oil), and 2020–2023 (COVID and post-pandemic inflation), with NBER bars sitting inside the recovered bear regions. The $K = 3$ and $K = 4$ models collapse two of their states (visible in those panels as essentially-flat lines near zero), indicating the 271-month panel and the two observation channels do not support richer regime structure.

4.2 Solver behaviour

The value-iteration and policy-iteration solvers agree exactly on the greedy policy and within 2.6×10^{-6} on the value function for every γ and every K we tested, as predicted by Proposition 1. Table 2 reports the iteration count.

4.3 Headline backtest

Table 3 and fig. 2 report the 2003–2026 backtest performance of the three policies under the headline setting ($K = 2$, $\gamma = 2$, $\lambda = 0.95$, 5 bps cost).

The QMDP policy beats the static benchmark on absolute return (9.9% vs. 7.4% CAGR) but *underperforms* on a risk-adjusted basis (Sharpe 0.72 vs. 0.81) and has a much deeper drawdown (−53% vs. −34%). The reason is visible in the policy table: under $\gamma = 2$ the underlying MDP’s optimal policy is $\pi^*(s) = 100\%$ stocks for both regimes. Hence the QMDP, no matter what the belief, recommends 100% stocks. The "regime overlay" is in this configuration a pure equity strategy. The myopic baseline dominates both because it implicitly tracks *trend*: a recent crash lowers the 12-month mean and pushes the allocation toward bonds.

Table 2: Solver iteration counts. VI uses the formula-sheet stopping criterion $\|V^{(n+1)} - V^{(n)}\|_\infty < \varepsilon(1 - \lambda)/(2\lambda)$.

K	VI ($\gamma=2$)	PI ($\gamma=2$)	VI ($\gamma=10$)	PI ($\gamma=10$)
2	125	2	125	2
3	96	2	96	2
4	96	2	96	2

Table 3: Headline backtest 2003–2026. Static 60/40 vs. QMDP vs. myopic one-step lookahead. All metrics annualised where applicable.

Policy	CAGR	Vol	Sharpe	Max DD	Calmar	Turnover
Static 60/40	7.35%	9.4%	0.81	−34.2%	0.22	0.000
QMDP ($\gamma = 2$)	9.90%	14.7%	0.72	−53.0%	0.19	0.000
Myopic 12-mo	13.99%	11.3%	1.22	−21.1%	0.66	0.236

4.4 Why QMDP collapses: CRRA risk-aversion sweep

Figure 3 reports the headline result of this paper. We sweep CRRA risk aversion $\gamma \in \{1, 2, 3, 5, 8, 10, 15, 20\}$ and re-solve the MDP for each. At $\gamma \leq 3$ the optimal policy is “100 % stocks” in every regime; $\gamma = 5$ shifts to 80/20; $\gamma = 8$ to 40/60; and $\gamma \geq 15$ to 20/80. Crucially, $\pi^*(s_{\text{bull}})$ equals $\pi^*(s_{\text{bear}})$ at *every* γ : the optimal policy does *not* differ across regimes; it merely shifts with the agent’s risk-tolerance level. The QMDP Sharpe ratio surpasses the static-60/40 benchmark (0.81) starting at $\gamma = 8$ and saturates near 0.89 at $\gamma \geq 15$.

Table 4 makes this concrete. Within each column the policy is constant across rows (regime-insensitive), but it shifts *column to column* as γ rises.

5 Conclusions

We built and stress-tested a POMDP-based regime overlay for stock-bond allocation. The pipeline is correct (VI and PI agree numerically; the 2-state HMM cleanly recovers the major stress episodes; the QMDP control is well-defined). At the same time, the headline backtest reveals a qualitative finding worth taking seriously: *the regime signal exists in our observations but the regime action does not*, in the sense that the optimal CRRA policy is the same in bull and bear regimes for every level of risk aversion we tested. The Sharpe gain from raising γ is a “second-moment” effect: more risk aversion reduces equity exposure across-the-board and therefore reduces drawdowns and volatility, but it does not exploit the regime label.

Implications

For a practitioner considering a regime-aware overlay, our results give three concrete cautions. First, two macro observables (VIX and the term spread) are not sufficient to differentiate the per-regime conditional expected returns enough to drive a regime-dependent policy. Second, log utility is too risk-tolerant: any case-study claiming gains from regime switching under log utility is almost certainly capturing *higher mean exposure* rather than *regime-aware tilting*. Third, the most useful single comparator is not the static 60/40 but a *myopic trend-following baseline*, which captures

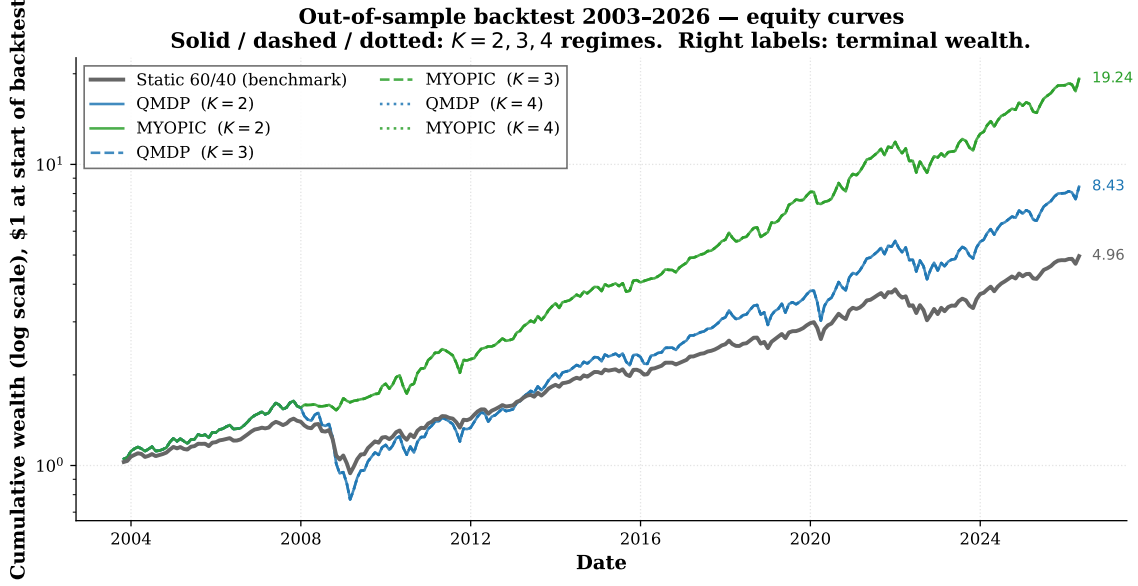


Figure 2: Cumulative wealth (log scale) from \$1 invested at the start of the backtest. Solid, dashed and dotted variants correspond to $K = 2, 3, 4$ regime configurations of the same QMDP and myopic policies; the static 60/40 benchmark is the heavier gray line. Terminal-wealth values are annotated to the right of each curve. Note that the QMDP equity curves overlap almost perfectly across K : the underlying MDP collapses to a 100%-stocks allocation under CRRA $\gamma = 2$ regardless of regime count, so the regime overlay reduces to a pure-equity strategy. The myopic policy, which uses a 12-month trailing return, dominates because it captures trend whenever stocks slip below bonds.

Table 4: Optimal stock/bond split $\pi^*(s)$ as a function of regime s and CRRA γ . Within columns the two regimes give the *same* action: the regime label does not change the policy. The policy shifts *across* columns as γ rises.

Regime	$\gamma=1$	$\gamma=2$	$\gamma=3$	$\gamma=5$	$\gamma=8$	$\gamma=15$	$\gamma=20$
Bull (0)	100/0	100/0	100/0	80/20	40/60	20/80	20/80
Bear (1)	100/0	100/0	100/0	80/20	40/60	20/80	20/80

recent first-moment information without any latent-state machinery and was the highest-Sharpe policy we tested.

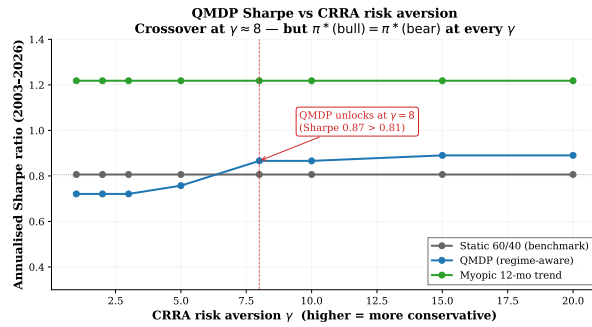
Extensions

We see three high-value extensions.

(i) *Richer observations.* The proposal called for ICE BofA HY OAS as a third observation channel. Including HY OAS would reduce the spectral overlap of the bull and bear emissions, plausibly producing regime-dependent $\pi^*(s)$. Implementation requires a registered FRED API key, which bypasses the public CSV truncation.

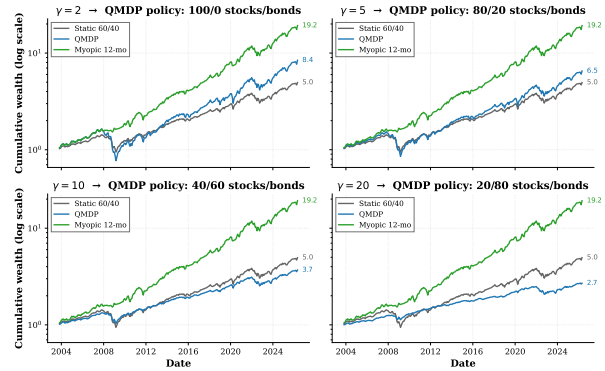
(ii) *Sharpe-style or CVaR-penalised reward.* A reward of the form $R(s, \mathbf{a}) = \mathbb{E}[\mathbf{a}^\top \mathbf{r} \mid s] - \kappa \text{Var}[\mathbf{a}^\top \mathbf{r} \mid s]$ explicitly penalises bear-regime volatility and is more likely to produce a regime-dependent policy. This is a one-line change in our reward module.

(iii) *Belief-space value iteration.* QMDP is the simplest belief-space policy. Point-based value iteration [8] maintains an explicit α -vector representation and would, in principle, exploit infor-



(a) Annualised Sharpe vs. γ .

Equity curves under varying CRRA risk aversion γ — higher γ shifts QMDP toward bonds across-the-board



(b) Equity curves at $\gamma \in \{2, 5, 10, 20\}$.

Figure 3: CRRA risk-aversion sweep. Panel (a) plots the annualised Sharpe ratio of each policy as γ varies from 1 to 20. The static-60/40 and myopic baselines are flat by construction. The QMDP Sharpe is below the static benchmark for $\gamma \leq 5$ and crosses above it at $\gamma = 8$, reaching 0.89 by $\gamma \geq 15$. Panel (b) shows the corresponding backtest equity curves at four representative γ values. Each panel title states the underlying MDP’s optimal stock/bond split, illustrating that higher risk aversion shifts the QMDP policy bondward across the board, but never differentiates the bull and bear regimes.

mation value (the action of *learning* the regime). On a 2-state model, the belief simplex is a 1-D segment, so PBVI is cheap and worth running as a diagnostic.

What this report demonstrates

End-to-end, the project assembles a hidden Markov chain over regimes, viewed through the lens of a Bayesian network, into a fully-observable Bellman recursion solved by value iteration and cross-checked against exact policy iteration, and wrapped in a belief-state controller via the QMDP approximation. The natural model-free alternative, a multi-armed-bandit or temporal-difference learner, is left to future work.

Reproducibility. All code, data-fetching, intermediate artifacts, and figures are committed to `takakhoo/ENG5177_Final_Project`. The end-to-end pipeline runs in under five minutes on a laptop.

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A Team contributions

Contributions reported following the report-guidelines template. Percentages indicate first-pass share; cross-review of all sections is by all four members.

Task	Dario	Even	Kyle (KDLL)	Taka
Defined the problem	25%	25%	25%	25%
Collected data	25%	25%	25%	25%
Performed data analysis	30%	20%	30%	20%
Built model of uncertainty (HMM)	70%	0%	20%	10%
Built model of decision (POMDP)	10%	30%	10%	50%
Developed solution technique (VI/PI/QMDP)	10%	60%	20%	10%
Drafted initial report	20%	20%	20%	40%
Revised the report	25%	25%	25%	25%
Drafted initial slides	25%	25%	25%	25%
Revised the slides	25%	25%	25%	25%